

Impact-Protection Lander Structure

Design a lightweight protective structure that limits damage to a payload during a controlled drop.

Advanced	2-3 class periods	Engineering Challenge
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Engineering Context

A small sensor package must survive a hard landing. Your team will design a structure that manages impact energy without simply surrounding the payload with excessive material.

What You Will Practice

- Apply energy absorption concepts
- Design controlled deformation
- Use iterative drop testing
- Evaluate protection versus mass

What You Need

- Small payload or raw egg substitute
- Paper, straws, foam, cardboard, tape, or approved materials
- Mass scale
- Drop test area
- Engineering notebook

Student Instructions

- 1 Record payload dimensions, mass, and failure risks.
- 2 Sketch two concepts using crumple zones, suspension, landing legs, or protective frames.
- 3 Select one concept and define a maximum structure mass.
- 4 Build the first prototype and photograph it before testing.
- 5 Conduct a low-height drop, inspect, and revise.
- 6 Increase to the official height only after passing the lower test.
- 7 Record damage, orientation, and structural deformation.
- 8 Calculate total system mass and summarize protection performance.

Engineering Requirements

- Payload must remain contained
- Structure must fit within the assigned envelope
- Mass must remain below the unit limit
- No loose material may scatter during testing
- Complete at least two test-and-revision cycles

Constraints & Safety

- Use only the designated drop zone
- No throwing or launching
- Wear eye protection if required
- Retrieve the device only after the area is clear

What You Submit

- Two concept sketches
- Prototype photos
- Test log
- Mass record
- Damage assessment
- Final redesign explanation

Reflection Questions

- 1 Which feature absorbed the most energy?
- 2 Did the structure protect by staying rigid or deforming?
- 3 How did landing orientation affect results?
- 4 What would change for a higher drop speed?

Engineering Tip Plan where the structure should deform; accidental deformation may transfer load directly to the payload.	Common Mistake Adding soft material everywhere can increase mass without controlling the load path.
Stretch Goal Improve survival height by 25% without increasing mass.	Career Connection Crashworthiness engineers design landing gear, energy absorbers, seats, and protective structures for aircraft and spacecraft.